

PRACTICAL 11

AIM

Define security and quality aspects of the identified module.

Objective

1. To identify potential security vulnerabilities in the PEFER health management system.
2. To establish quality assurance measures for the system's functionality, accuracy, and reliability.
3. To ensure compliance with data protection regulations (e.g., GDPR, HIPAA).
4. To implement best practices for secure data handling and system performance.

Prerequisites

1. Basic understanding of AI/ML concepts.
2. Familiarity with healthcare data security standards.
3. Knowledge of software development lifecycle (SDLC).
4. Access to a PEFER dataset (simulated or real, anonymized).

Materials Required

1. Laptop/PC with Python installed.
2. AI/ML libraries (e.g., TensorFlow, PyTorch, Scikit-learn).
3. Database management system (e.g., MySQL, PostgreSQL).
4. Security testing tools (e.g., OWASP ZAP, Burp Suite).
5. Data encryption tools (e.g., AES, RSA).

6. Quality assurance tools (e.g., Selenium, JUnit).

Procedure

Step 1: Identify the Module

1. Define the module to be evaluated (e.g., PEFR data collection, AI-based prediction, user dashboard).
2. Document the module's functionality, inputs, outputs, and dependencies.

Step 2: Security Aspects

1. Data Security:

- Ensure all PEFR data is encrypted during transmission (e.g., HTTPS, TLS).
- Implement encryption for data at rest (e.g., AES-256).
- Use secure authentication mechanisms (e.g., OAuth, multi-factor authentication).

2. Access Control:

- Define role-based access control (RBAC) for users (e.g., patients, doctors, admins).
- Restrict access to sensitive data based on user roles.

3. Vulnerability Assessment:

- Perform penetration testing using tools like OWASP ZAP.
- Identify and mitigate common vulnerabilities (e.g., SQL injection, XSS).

4. Compliance:

- Ensure compliance with healthcare regulations (e.g., GDPR, HIPAA).
- Conduct regular audits to verify compliance.

Step 3: Quality Aspects

1. Functional Testing:

- Verify that the module performs its intended functions accurately.
- Test edge cases (e.g., invalid PEFR values, missing data).

2. Performance Testing:

- Evaluate the system's response time under different loads.
- Ensure the AI model provides predictions within acceptable time limits.

3. Accuracy Testing:

- Validate the AI model's predictions against a labeled dataset.

- Calculate performance metrics (e.g., accuracy, precision, recall).

4. Usability Testing:

- Ensure the user interface is intuitive and accessible.
- Gather feedback from end-users (e.g., patients, doctors).

Step 4: Documentation and Reporting

1. Document all security and quality measures implemented.
2. Create a report summarizing the findings, including:
 - Identified vulnerabilities and their mitigation strategies.
 - Quality metrics (e.g., accuracy, response time).
 - Recommendations for improvement.

Expected Outcomes

1. A secure and reliable PEFr health management module.
2. Documentation of security and quality measures.
3. A report highlighting vulnerabilities, quality metrics, and recommendations.